

Predictive steering wins where the loss is declared

Re-run after an audit of the evaluator, on 25 000 paired simulated days. The controllers now beat the simple rule — but only for capacity loss that is announced in advance, and the boundary is as important as the gain.

+24.0%

OVERLOAD REMOVED
ON UNSEEN SEEDS

0 / 473

REGRESSIONS ON
DECLARED EVENTS

36 / 160

CONFIGURATIONS PASSED
EVERY GATE

3 / 4

SURVIVED FRESH-SEED
VALIDATION

25 000

PAIRED SIMULATED
DAYS

WHAT THE CONTROLLERS NOW WIN

On declared maintenance the guarded pre-drain removes 55.2% of uplink overload, and on maintenance followed by a stadium-scale surge, 52.2%. Across 473 pairs of those two families it did not lose a single one.

Cohort MPC clears every gate too, at 33.1% and 32.4%, with a tenth of the collateral harm and a tenth of the routing churn.

WHAT IT STILL COSTS

When maintenance is announced and then a second, unannounced brownout lands on a box the controller has already drained onto, pre-drain loses in 30.5% of cases. The family is still net positive, but the worst cases are severe.

MPC cuts that exposure to 6.6% of cases. That is the trade C-DOT chooses between.

Five caveats that bound every number in this deck

None of these are disclaimers added afterwards. Each one changes how a figure on a later slide should be read.

CAVEAT 1

This is a shadow controller in simulation.

No policy touched live traffic. Every record in the project is labelled synthetic and none of it is calibrated to real C-DOT capacity figures.

CAVEAT 2

Only about 3% of load is controllable.

The lever is where NEW sessions are placed. Sessions already established are never migrated, so a controller can only steer the margin.

CAVEAT 3

Gains are for announced capacity loss only.

With no declared event the controller publishes bit-exact static output. Both pure-surprise families are deliberate ties, verified byte for byte.

CAVEAT 4

One stress family produced no measurable stress.

An unannounced demand surge never overloaded any box, even at 4x arrivals across a zone. That family scores a guaranteed zero, so the headline is conservative.

CAVEAT 5

Chosen on one seed pool, judged on another.

The headline numbers come from a pool never inspected before the candidates were frozen. Screening numbers are 3-4 points higher and are not the claim.

HOW TO READ THE OPTIMIZER SECTION

Experiments 1-6 are the earlier campaign.

They remain on the record for method and for the negative results. Experiment 7 supersedes their conclusion.

Re-running the campaign after auditing the evaluator

The earlier campaign concluded no controller could beat the simple rule. An audit found three defects — all in the scoring code and the scenario generator, none in the controllers themselves. Fixing them changed the answer.

DEFECT 1 · UNITS

The optimiser and the scoreboard disagreed

The simulator scores overload as load / capacity - 1, a relative quantity. The safety guard and the pre-drain solver both worked in absolute Mbps. On a box cut to a tenth of its envelope those differ tenfold, so the controller undervalued exactly the action it exists to take.

DEFECT 2 · METRIC

One stress family scored infinity

An unavailable box has zero capacity, so those pairs scored infinity for both controllers — an exact tie that still failed a finite-metric check on all 160 configurations. That single check made the contract unpassable before any controller ran.

DEFECT 3 · SIZING

The MPC horizon was set in windows, not hours

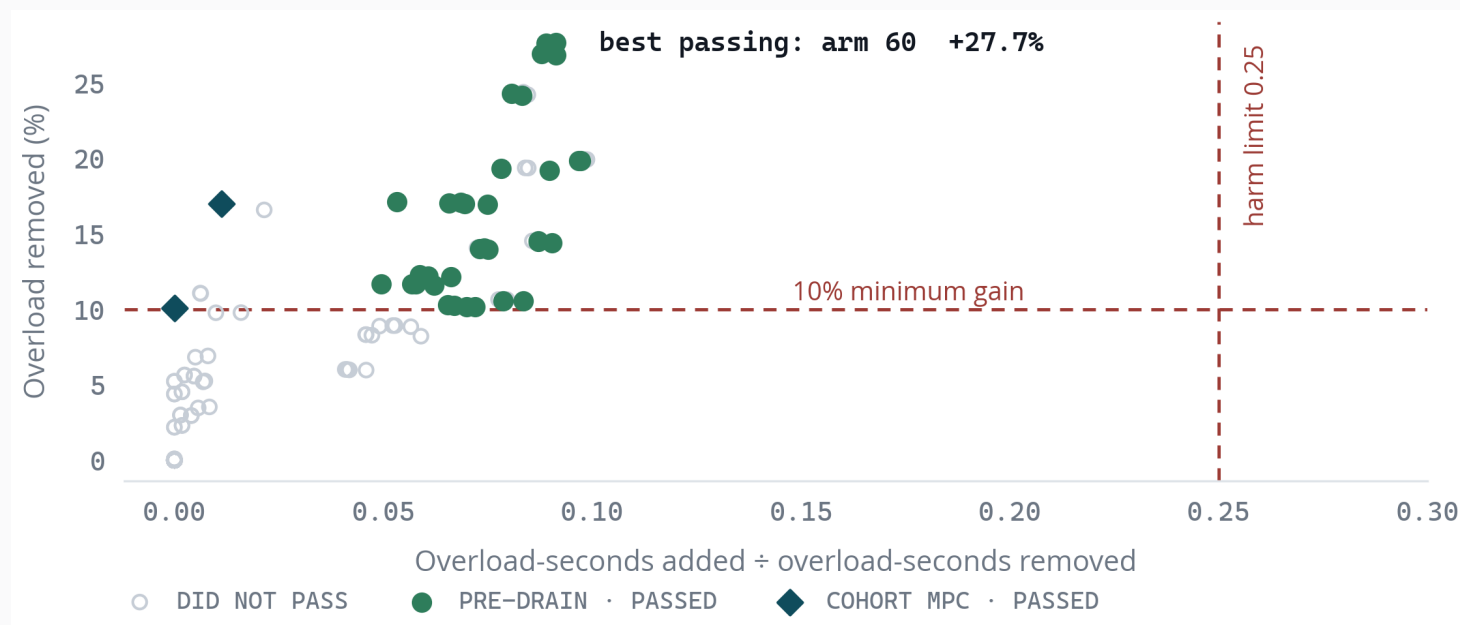
Halving the control cadence quintupled the optimisation, so every 2-minute configuration hit a 2-second solver budget that the acceptance rules allowed to be 120 seconds. The reported cadence finding was measuring a sizing bug.

WHAT CHANGED AS A RESULT

Configurations passing every gate: 0 of 160 to 36 of 160. Worst stress family: -1.7% to +24.2%. MPC solver timeouts: 40% at 10-minute cadence and 100% at 2-minute, now zero.

160 configurations, 20 000 paired simulated days

Every configuration is one point. To pass it must clear a 10% minimum gain and add no more than a quarter of the overload it removes — plus nine further checks.



■ 36 of 160 cleared every check

against 0 of 160 in the earlier campaign, on the same simulator and the same 24-hour scenarios.

■ Pre-drain reaches further

the strongest passing configuration removes 27.7% of overload while adding 9% of what it removes.

■ MPC sits low and left

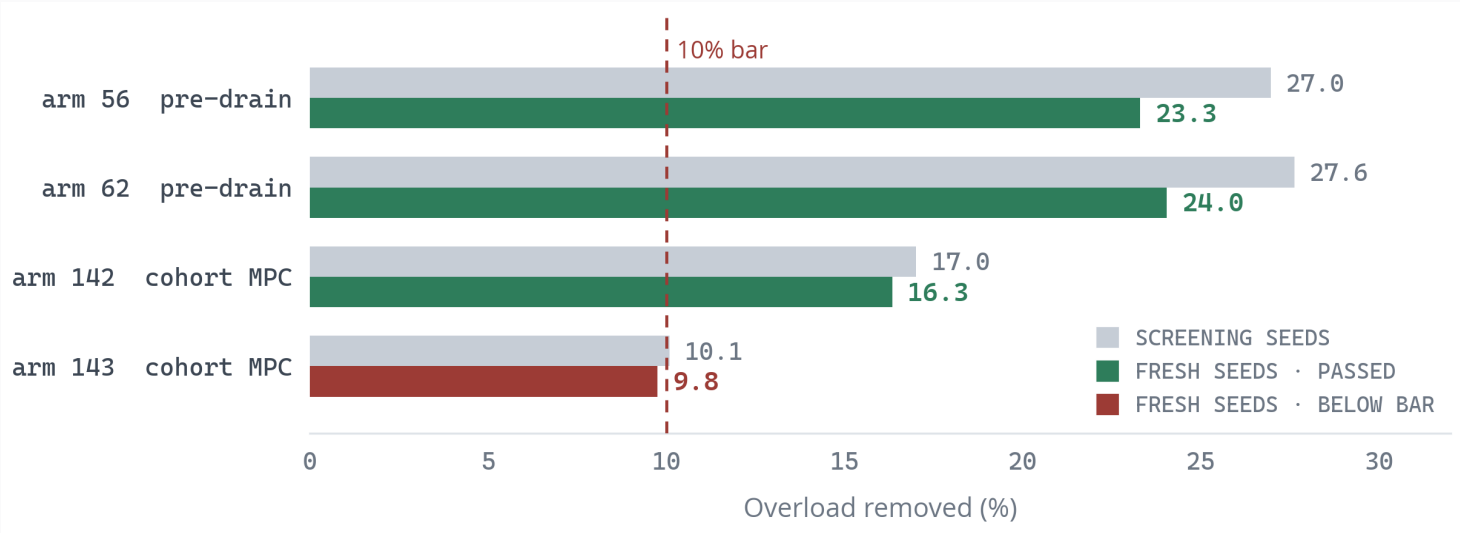
less gain, but collateral harm near zero — the conservative end of the same frontier.

THE SHAPE OF THE CAMPAIGN

Each configuration ran the same 125 paired days: five stress families x 25 seeds, one per CPU. 20 000 paired 24-hour simulations in total, 160 of 160 arms valid, zero non-finite results.

Frozen candidates, on seeds never inspected

Configurations that look good on the data that chose them are not evidence. The four leaders were frozen by a rule written down in advance — ranked on the lower confidence bound, not the headline — then re-run on a separate seed pool.



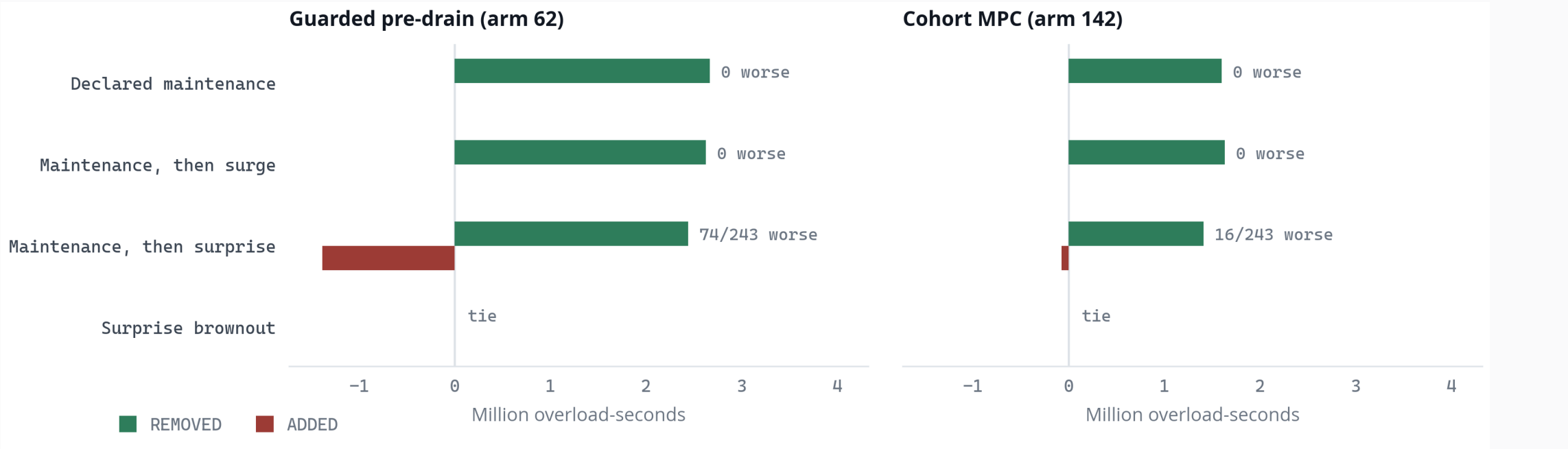
- **Three of four held every gate**
arm 143 fell just below the 10% bar at 9.8% and was not promoted.
- **The drop is 3-4 points, not a collapse**
that gap is the selection bias being removed. A candidate that fell apart here would have been a screening artifact; none did.
- **Seeds are provably disjoint**
from the screening pool, from every earlier campaign, and from the protected test set.

WHY THE SECOND POOL EXISTS

1 250 fresh paired days per candidate, 250 per stress family, drawn from seeds 81 000 and above — disjoint from the 80 000-80 124 screening pool, from every earlier campaign, and from the protected test set.

Three states, not two

A claim that predictive steering is simply better would not survive scrutiny, and it is not what the data says. Bars right of the line are overload removed; bars left are overload the controller added. Both panels share one scale.



Declared loss — wins, never loses

473 informative pairs, 0 regressions. Median case removes 72% (pre-drain) and 40% (MPC) of uplink overload.

Pure surprise — bit-exact ties

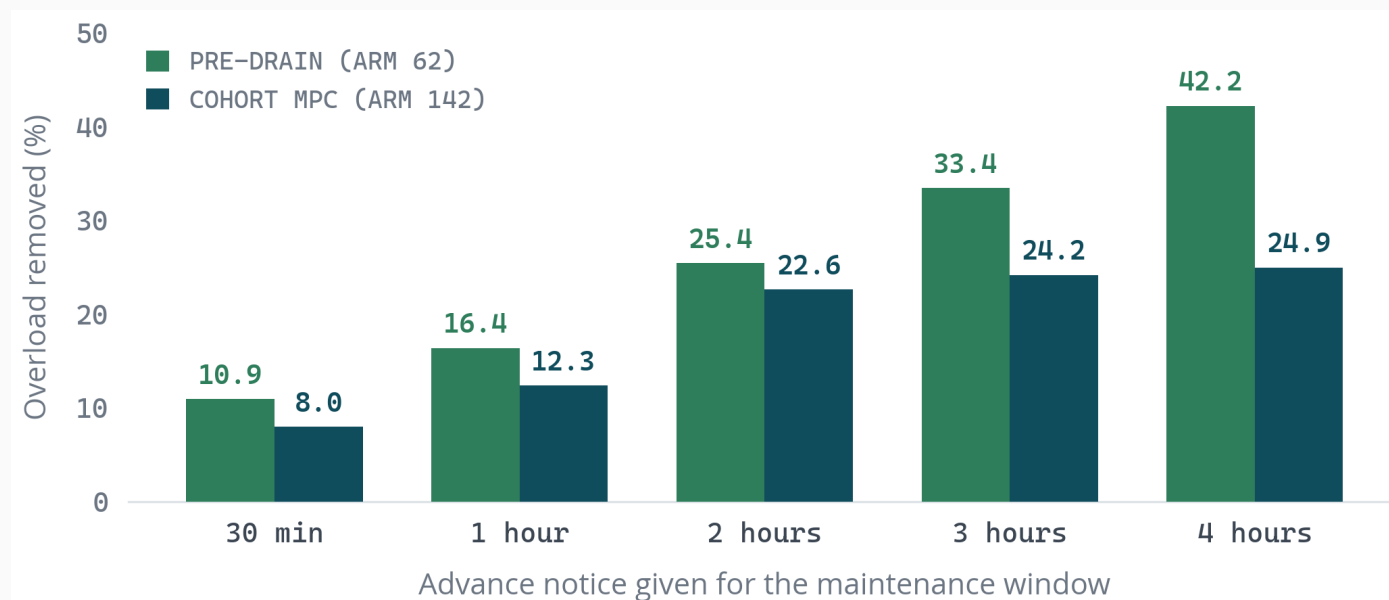
With no declared event the published policy is byte-for-byte identical to the simple rule. Verified, not assumed.

Declared loss, then a surprise — can lose

Pre-drain regresses in 74 of 243 pairs; MPC in 16 of 243. Still net positive, but this is the real exposure.

The resource that actually matters

Gain scales with how much warning the maintenance window carries. Notice is a property of the scenario, drawn from its seed, so every configuration faced the same schedule.



■ Four hours is worth 4x thirty minutes

pre-drain removes 42.2% of overload with four hours' warning and 10.9% with thirty minutes.

■ MPC saturates around 24%

its two-hour planning horizon cannot exploit three or four hours of notice; pre-drain keeps climbing.

■ This is a scheduling lever, not a model

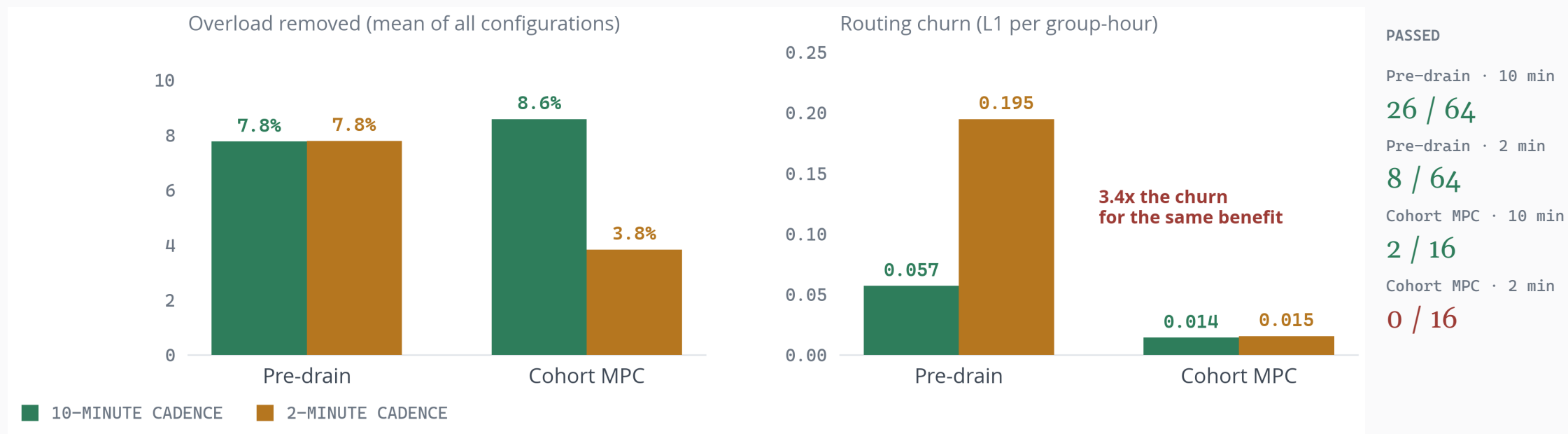
the single cheapest way to increase the benefit is to announce maintenance earlier.

WHAT THIS MEANS OPERATIONALLY

Announcing a maintenance window three hours ahead instead of thirty minutes roughly triples the overload the controller can remove. That is a change to scheduling practice, not to any model.

Two minutes buys churn, not benefit

Running the controller every 2 minutes instead of every 10 was tested across all 160 configurations, with the optimisation held to the same size so cadence trades lookahead against reactivity rather than against solver feasibility.



THE SAME CONFIGURATION, RUN AT BOTH CADENCES

Arm 60 at 10 minutes removes 27.67% of overload with 0.192 churn and passes every check. Arm 124 — identical apart from cadence — removes 27.69% with 0.702 churn and fails on churn alone. Faster control bought two hundredths of a point for 3.7x the routing change.